

Emre Tiryaki

Software Engineering Student – Backend Developer

📍 Samsun, Turkey

✉ tiryakiemre18@gmail.com

🌐 in/emre-tiryaki

☎ +90 544 319 52 15

🌐 emre-tiryaki.dev

🔄 emre-tiryaki



Profile Summary

Software engineering student focused on distributed systems, microservice architectures, and developing scalable backend applications. Builds applications with Go, Java, TypeScript, and Rust; designs modular, testable, and maintainable services using NestJS, Spring Boot, Gin, GraphQL, and RabbitMQ. Adopts modern software development practices by actively utilizing Docker, Linux, and CI/CD pipelines.

Education

Inonu University

Software Engineering (B.Sc.)

Malatya, Turkey

2023 – 2027 (Expected)

- 4th-year student; focus areas: backend development, DevOps, distributed systems.

Technical Skills

Programming Languages	TypeScript, JavaScript, Go, Rust, Java
Backend	Gin, Express, Axum, Fastify, NestJS, Spring Boot, GraphQL, REST API, RabbitMQ
Databases	PostgreSQL, MySQL, MongoDB
Distributed Systems	Asynchronous messaging, inter-service communication, scalable architecture design
DevOps / Infrastructure	Docker, CI/CD (Continuous Integration / Continuous Deployment), Linux, Git

Experience

Avikon – Backend Developer Intern

August 2026 – Present

- Developed and maintained Fastify-based backend services.
- Designed and implemented a GraphQL API layer using the Mercurius library.
- Built an asynchronous inter-service messaging infrastructure utilizing RabbitMQ.

Projects

Autonomous Financial Assistant (OFA) – Smart Payment Routing and Problem Solver

July 2026

Moka United AI Ideathon & Hackathon

JavaScript, Express, Python, Machine Learning, React

- Developed a machine learning-based algorithm to select the most optimal acquirer bank for transactions.
- Designed an AI agent architecture that takes autonomous actions to resolve system issues without requiring human intervention.

OKUR – AI Exam Grader

May 2026

Inufest 2026 Competition

TypeScript, NestJS, Python, Machine Learning, React

- Built a system that automatically generates exam papers in LaTeX format.
- Implemented an image processing module that crops and transcribes student answers from scanned physical papers.
- Developed an AI model that semantically compares student answers with the instructor's answer key to automate grading.

Cosmic Data Processing Pipeline – Big Data Processing Engine

March 2026

TUA Astro Hackathon

Python, FastAPI, React

- Built a component to generate synthetic cosmic data and transmit it directly to the processing engine.
- Implemented an end-to-end data processing pipeline utilizing real-time digital signal processing (DSP) via NumPy and SciPy vectorization to clean cosmic radiation-induced signal degradation.
- Achieved 90%+ autonomous anomaly detection accuracy using LSTM Autoencoder models.

HEGS News – Gamified News Platform

October 2025

Anadolu Agency Media Technologies Hackathon

JavaScript, Java, Spring Boot, Python, FastAPI

- Scraped news data from the Anadolu Agency website and presented it through a clean, user-friendly interface.
- Integrated an AI module to provide users with dynamically generated summaries of current news.

Languages

Turkish Native

English Professional working proficiency

Independent Work

- Enhanced Bash scripting skills on Arch Linux, utilizing it as a daily driver operating system.
- Developed **project-toolkit**, a bash script automating project creation and configuration workflows.
- Created **auto-readme-generator** (Rust/Axum backend, React/JavaScript frontend), a tool that scans repositories to automatically generate README files using user-specified AI models.
- Built **pdf2markdown**, a Python CLI tool that converts PDF files to Markdown concurrently via locally hosted Ollama models.
- Developed **RepoGrab**, a Go-based TUI (Terminal User Interface) tool allowing users to download individual files from GitHub and GitLab without cloning the entire repository.